Bits	Description	Type	Reset
11:0	**BASE**: Physical address base for this virtual address range, in units of 4 kiB (one flash sector). Taking a 24-bit virtual address, firstly bits 23:22 (the two MSBs) are masked to zero, and then BASE is added to bits 23:12 (the upper 12 bits) to form the physical address. Translation wraps on a 16 MiB boundary.	RW	0x800

QMI: ATRANS3, ATRANS7 Registers

Offsets: 0x40, 0x50

Description

Configure address translation for XIP virtual addresses 0xc00000 through 0xffffff (a 4 MiB window starting at +12 MiB).

Address translation allows a program image to be executed in place at multiple physical flash addresses (for example, a double-buffered flash image for over-the-air updates), without the overhead of position-independent code.

At reset, the address translation registers are initialised to an identity mapping, so that they can be ignored if address translation is not required.

Note that the XIP cache is fully virtually addressed, so a cache flush is required after changing the address translation.

Table 1305. ATRANS3, ATRANS7 Registers

Bits	Description	Type	Reset
31:27	Reserved.	-	-
26:16	**SIZE**: Translation aperture size for this virtual address range, in units of 4 kiB (one flash sector). Bits 21:12 of the virtual address are compared to SIZE. Offsets greater than SIZE return a bus error, and do not cause a QSPI access.	RW	0x400
15:12	Reserved.	-	-
11:0	**BASE**: Physical address base for this virtual address range, in units of 4 kiB (one flash sector). Taking a 24-bit virtual address, firstly bits 23:22 (the two MSBs) are masked to zero, and then BASE is added to bits 23:12 (the upper 12 bits) to form the physical address. Translation wraps on a 16 MiB boundary.	RW	0xc00

12.15. System Control Registers

These registers are not associated with any particular peripheral. They control, or provide information about, system-level hardware such as the bus fabric. This is also where chip identification information such as the JEDEC IDCODE is provided in a software-accessible manner.

12.15.1. SYSINFO

12.15.1.1. Overview

The sysinfo block contains system information. The first register contains the Chip ID, which allows the programmer to know which version of the chip software is running on. The second register indicates which package configuration is

used (QFN-60 or QFN-80). The third register will always read as 1.

12.15.1.2. List of registers

The sysinfo registers start at a base address of `0x40000000` (defined as SYSINFO_BASE in SDK).

Table 1306. List of SYSINFO registers

Offset	Name	Info
0x00	CHIP_ID	JEDEC JEP-106 compliant chip identifier.
0x04	PACKAGE_SEL	Package selection indicator, 0 = QFN80, 1 = QFN60
0x08	PLATFORM	Platform register. Allows software to know what environment it is running in during pre-production development. Post-production, the PLATFORM is always ASIC, non-SIM.
0x14	GITREF_RP2350	Git hash of the chip source. Used to identify chip version.

SYSINFO: CHIP_ID Register

Offset: 0x00

Description

JEDEC JEP-106 compliant chip identifier.

Table 1307. CHIP_ID Register

Bits	Description	Type	Reset
31:28	**REVISION**	RO	-
27:12	**PART**	RO	-
11:1	**MANUFACTURER**	RO	-
0	**STOP_BIT**	RO	0x1

SYSINFO: PACKAGE_SEL Register

Offset: 0x04

Table 1308. PACKAGE_SEL Register

Bits	Description	Type	Reset
31:1	Reserved.	-	-
0	Package selection indicator, 0 = QFN80, 1 = QFN60	RO	0x0

SYSINFO: PLATFORM Register

Offset: 0x08

Description

Platform register. Allows software to know what environment it is running in during pre-production development. Post-production, the PLATFORM is always ASIC, non-SIM.

Table 1309. PLATFORM Register

Bits	Description	Type	Reset
31:5	Reserved.	-	-
4	**GATESIM**	RO	-
3	**BATCHSIM**	RO	-
2	**HDLSIM**	RO	-
1	**ASIC**	RO	-

Bits	Description	Type	Reset
0	**FPGA**	RO	-

SYSINFO: GITREF_RP2350 Register

Offset: 0x14

Table 1310. GITREF_RP2350 Register

Bits	Description	Type	Reset
31:0	Git hash of the chip source. Used to identify chip version.	RO	-

12.15.2. SYSCFG

12.15.2.1. Overview

The system config block controls miscellaneous chip settings, including:

- Check debug halt status of both cores
- Processor GPIO input synchroniser control (set to 1 to allow input synchroniser bypassing to reduce latency for synchronous clocks)
- SWD interface control from inside the chip (allows one core to debug another, which may make debug connectivity easier)
- State-retaining memory power down (SRAM periphery can be powered down when not in use to save a small amount of power)
  - when powered down in this way, power is still applied to the SRAM storage array; use the Power Manager (Chapter 6) to completely remove power
- Additional controls found in the AUXCTRL register

12.15.2.2. Changes from RP2040

- Moved the NMI mask to per-core registers in the EPPB (Section 3.7.5.1). The new registers reset on a processor warm reset, which avoids issues with NMIs asserting during the bootrom early boot process.
- Expanded MEMPOWERDOWN to cover new memory banks
- Removed controls from DBGFORCE to account for the new single-DP debug topology

12.15.2.3. List of registers

The system config registers start at a base address of `0x40008000` (defined as SYSCFG_BASE in SDK).

Table 1311. List of SYSCFG registers

Offset	Name	Info
0x00	PROC_CONFIG	Configuration for processors

Offset	Name	Info
0x04	PROC_IN_SYNC_BYPASS	For each bit, if 1, bypass the input synchronizer between that GPIO and the GPIO input register in the SIO. The input synchronizers should generally be unbypassed, to avoid injecting metastabilities into processors. If you're feeling brave, you can bypass to save two cycles of input latency. This register applies to GPIO 0…31.
0x08	PROC_IN_SYNC_BYPASS_HI	For each bit, if 1, bypass the input synchronizer between that GPIO and the GPIO input register in the SIO. The input synchronizers should generally be unbypassed, to avoid injecting metastabilities into processors. If you're feeling brave, you can bypass to save two cycles of input latency. This register applies to GPIO 32…47. USB GPIO 56..57 QSPI GPIO 58..63
0x0c	DBGFORCE	Directly control the chip SWD debug port
0x10	MEMPOWERDOWN	Control PD pins to memories. Set high to put memories to a low power state. In this state the memories will retain contents but not be accessible Use with caution
0x14	AUXCTRL	Auxiliary system control register

SYSCFG: PROC_CONFIG Register

Offset: 0x00

Description

Configuration for processors

Table 1312. PROC_CONFIG Register

Bits	Description	Type	Reset
31:2	Reserved.	-	-
1	**PROC1_HALTED**: Indication that proc1 has halted	RO	0x0
0	**PROC0_HALTED**: Indication that proc0 has halted	RO	0x0

SYSCFG: PROC_IN_SYNC_BYPASS Register

Offset: 0x04

Description

For each bit, if 1, bypass the input synchronizer between that GPIO
and the GPIO input register in the SIO. The input synchronizers should
generally be unbypassed, to avoid injecting metastabilities into processors.
If you're feeling brave, you can bypass to save two cycles of input
latency. This register applies to GPIO 0…31.

Table 1313. PROC_IN_SYNC_BYPASS Register

Bits	Description	Type	Reset
31:0	**GPIO**	RW	0x00000000

SYSCFG: PROC_IN_SYNC_BYPASS_HI Register

Offset: 0x08

Description

For each bit, if 1, bypass the input synchronizer between that GPIO and the GPIO input register in the SIO. The input synchronizers should generally be unbypassed, to avoid injecting metastabilities into processors. If you're feeling brave, you can bypass to save two cycles of input latency. This register applies to GPIO 32…47. USB GPIO 56..57 QSPI GPIO 58..63

Table 1314. PROC_IN_SYNC_BYPASS_HI Register

Bits	Description	Type	Reset
31:28	**QSPI_SD**	RW	0x0
27	**QSPI_CSN**	RW	0x0
26	**QSPI_SCK**	RW	0x0
25	**USB_DM**	RW	0x0
24	**USB_DP**	RW	0x0
23:16	Reserved.	-	-
15:0	**GPIO**	RW	0x0000

SYSCFG: DBGFORCE Register

Offset: 0x0c

Description

Directly control the chip SWD debug port

Table 1315. DBGFORCE Register

Bits	Description	Type	Reset
31:4	Reserved.	-	-
3	**ATTACH**: Attach chip debug port to syscfg controls, and disconnect it from external SWD pads.	RW	0x0
2	**SWCLK**: Directly drive SWCLK, if ATTACH is set	RW	0x1
1	**SWDI**: Directly drive SWDIO input, if ATTACH is set	RW	0x1
0	**SWDO**: Observe the value of SWDIO output.	RO	-

SYSCFG: MEMPOWERDOWN Register

Offset: 0x10

Description

Control PD pins to memories.
Set high to put memories to a low power state. In this state the memories will retain contents but not be accessible
Use with caution

Table 1316. MEMPOWERDOWN Register

Bits	Description	Type	Reset
31:13	Reserved.	-	-
12	**BOOTRAM**	RW	0x0

Bits	Description	Type	Reset
11	**ROM**	RW	0x0
10	**USB**	RW	0x0
9	**SRAM9**	RW	0x0
8	**SRAM8**	RW	0x0
7	**SRAM7**	RW	0x0
6	**SRAM6**	RW	0x0
5	**SRAM5**	RW	0x0
4	**SRAM4**	RW	0x0
3	**SRAM3**	RW	0x0
2	**SRAM2**	RW	0x0
1	**SRAM1**	RW	0x0
0	**SRAM0**	RW	0x0

SYSCFG: AUXCTRL Register

Offset: 0x14

Description

Auxiliary system control register

Table 1317. AUXCTRL Register

Bits	Description	Type	Reset
31:8	Reserved.	-	-
7:0	* Bits 7:3: Reserved * Bit 2: Set to mask OTP power analogue power supply detection from resetting OTP controller and PSM * Bit 1: When clear, the LPOSC output is XORed into the TRNG ROSC output as an additional, uncorrelated entropy source. When set, this behaviour is disabled. * Bit 0: Force POWMAN clock to switch to LPOSC, by asserting its WDRESET input. This must be set before initiating a watchdog reset of the RSM from a stage that includes CLOCKS, if POWMAN is running from clk_ref at the point that the watchdog reset takes place. Otherwise, the short pulse generated on clk_ref by the reset of the CLOCKS block may affect POWMAN register state.	RW	0x00

12.15.3. TBMAN

TBMAN refers to the testbench manager, used during chip development simulations to verify the design. During these simulations TBMAN allows software running on RP2350 to control the testbench and simulation environment. On the real chip, it has no effect other than providing a single `PLATFORM` register that indicates that this is the real chip. This `PLATFORM` functionality is duplicated in the sysinfo (Section 12.15.1) registers.

12.15.3.1. List of registers

The TBMAN registers start at a base address of `0x40160000` (defined as TBMAN_BASE in SDK).

Table 1318. List of TBMAN registers

Offset	Name	Info
0x0	PLATFORM	Indicates the type of platform in use

TBMAN: PLATFORM Register

Offset: 0x0

Description

Indicates the type of platform in use

Table 1319. PLATFORM Register

Bits	Description	Type	Reset
31:3	Reserved.	-	-
2	**HDLSIM**: Indicates the platform is a simulation	RO	0x0
1	**FPGA**: Indicates the platform is an FPGA	RO	0x0
0	**ASIC**: Indicates the platform is an ASIC	RO	0x1

12.15.4. BUSCTRL

This block provides basic controls and monitoring for the system bus fabric.

12.15.4.1. Bus priority

RP2350 implements a dynamic bus priority scheme described in Section 2.1.1. The BUS_PRIORITY register implements the priority controls for this scheme.

12.15.4.2. Performance counters

There are four 24-bit counters, each of which can subscribe to a single performance event from the system bus fabric. Counters saturate at a value of all-ones: the counter stops incrementing when it reaches its maximum value, rather than wrapping to zero.

The performance counters are initially disabled: you must write `1` to PERFCTR_EN before the counters begin to increment. Write any value to a counter to clear the counter to zero in before running a profiled section of code, and enable the counters immediately before entering the profiled section. Disable the counters again immediately upon leaving the profiled section. The counters do not support arbitrary writes: they only count up from zero.

Write to a performance event selector register PERFSEL0 through PERFSEL3 to select the performance event which increments the corresponding counter, PERFCTR0 through PERFCTR3.

For each of the seventeen downstream bus ports on the main system AHB5 crossbar shown in Figure 5, there are four types of event which the performance counters detect. These events do not distinguish reads from writes, but they do distinguish different types of bus stall, which can be helpful when diagnosing performance issues. The types of event are:

Access

Increment when any access completes on this downstream port.

Contested access

Increment when any access completes on this downstream port which previously stalled due to the port being

occupied by another access. For example, if two managers access an initially idle port simultaneously, one will complete before the other. The access that completes first is said to not be contested, and does not increment this counter. The access that completes second (which was initially deferred due to the access from the other manager) *is* contested, and increments this counter when it completes.

Upstream-stalled cycle

Increment once per cycle while any manager experiences a stall on this port. This may be either due to arbitration with another manager (a contested access) or due to a stall on the downstream bus port, such as access to a slow peripheral. This is measured *at the port*, before leaving the main AHB5 crossbar.

Downstream-stalled cycle

Increment once per cycle while this port itself experiences a stall on the downstream bus. This indicates the peripheral or memory device itself being slow to respond, such as an XIP cache miss.

The first two event types listed above are the same as RP2040. The latter two are new for RP2350.

12.15.4.3. List of registers

The Bus Fabric registers start at a base address of `0x40068000` (defined as BUSCTRL_BASE in SDK).

Table 1320. List of BUSCTRL registers

Offset	Name	Info
0x00	BUS_PRIORITY	Set the priority of each master for bus arbitration.
0x04	BUS_PRIORITY_ACK	Bus priority acknowledge
0x08	PERFCTR_EN	Enable the performance counters. If 0, the performance counters do not increment. This can be used to precisely start/stop event sampling around the profiled section of code. The performance counters are initially disabled, to save energy.
0x0c	PERFCTR0	Bus fabric performance counter 0
0x10	PERFSEL0	Bus fabric performance event select for PERFCTR0
0x14	PERFCTR1	Bus fabric performance counter 1
0x18	PERFSEL1	Bus fabric performance event select for PERFCTR1
0x1c	PERFCTR2	Bus fabric performance counter 2
0x20	PERFSEL2	Bus fabric performance event select for PERFCTR2
0x24	PERFCTR3	Bus fabric performance counter 3
0x28	PERFSEL3	Bus fabric performance event select for PERFCTR3

BUSCTRL: BUS_PRIORITY Register

Offset: 0x00

Description

Set the priority of each master for bus arbitration.

Table 1321. BUS_PRIORITY Register

Bits	Description	Type	Reset
31:13	Reserved.	-	-
12	**DMA_W**: 0 - low priority, 1 - high priority	RW	0x0
11:9	Reserved.	-	-
8	**DMA_R**: 0 - low priority, 1 - high priority	RW	0x0

Bits	Description	Type	Reset
7:5	Reserved.	-	-
4	**PROC1**: 0 - low priority, 1 - high priority	RW	0x0
3:1	Reserved.	-	-
0	**PROC0**: 0 - low priority, 1 - high priority	RW	0x0

BUSCTRL: BUS_PRIORITY_ACK Register

Offset: 0x04

Description

Bus priority acknowledge

Table 1322. BUS_PRIORITY_ACK Register

Bits	Description	Type	Reset
31:1	Reserved.	-	-
0	Goes to 1 once all arbiters have registered the new global priority levels. Arbiters update their local priority when servicing a new nonsequential access. In normal circumstances this will happen almost immediately.	RO	0x0

BUSCTRL: PERFCTR_EN Register

Offset: 0x08

Table 1323. PERFCTR_EN Register

Bits	Description	Type	Reset
31:1	Reserved.	-	-
0	Enable the performance counters. If 0, the performance counters do not increment. This can be used to precisely start/stop event sampling around the profiled section of code. The performance counters are initially disabled, to save energy.	RW	0x0

BUSCTRL: PERFCTR0 Register

Offset: 0x0c

Description

Bus fabric performance counter 0

Table 1324. PERFCTR0 Register

Bits	Description	Type	Reset
31:24	Reserved.	-	-
23:0	Busfabric saturating performance counter 0 Count some event signal from the busfabric arbiters, if PERFCTR_EN is set. Write any value to clear. Select an event to count using PERFSEL0	WC	0x000000

BUSCTRL: PERFSEL0 Register

Offset: 0x10

Description

Bus fabric performance event select for PERFCTR0

Table 1325. PERFSEL0 Register

Bits	Description	Type	Reset
31:7	Reserved.	-	-
6:0	Select an event for PERFCTR0. For each downstream port of the main crossbar, four events are available: ACCESS, an access took place; ACCESS_CONTESTED, an access took place that previously stalled due to contention from other masters; STALL_DOWNSTREAM, count cycles where any master stalled due to a stall on the downstream bus; STALL_UPSTREAM, count cycles where any master stalled for any reason, including contention from other masters.	RW	0x1f
	Enumerated values:		
	0x00 → SIOB_PROC1_STALL_UPSTREAM		
	0x01 → SIOB_PROC1_STALL_DOWNSTREAM		
	0x02 → SIOB_PROC1_ACCESS_CONTESTED		
	0x03 → SIOB_PROC1_ACCESS		
	0x04 → SIOB_PROC0_STALL_UPSTREAM		
	0x05 → SIOB_PROC0_STALL_DOWNSTREAM		
	0x06 → SIOB_PROC0_ACCESS_CONTESTED		
	0x07 → SIOB_PROC0_ACCESS		
	0x08 → APB_STALL_UPSTREAM		
	0x09 → APB_STALL_DOWNSTREAM		
	0x0a → APB_ACCESS_CONTESTED		
	0x0b → APB_ACCESS		
	0x0c → FASTPERI_STALL_UPSTREAM		
	0x0d → FASTPERI_STALL_DOWNSTREAM		
	0x0e → FASTPERI_ACCESS_CONTESTED		
	0x0f → FASTPERI_ACCESS		
	0x10 → SRAM9_STALL_UPSTREAM		
	0x11 → SRAM9_STALL_DOWNSTREAM		
	0x12 → SRAM9_ACCESS_CONTESTED		
	0x13 → SRAM9_ACCESS		
	0x14 → SRAM8_STALL_UPSTREAM		
	0x15 → SRAM8_STALL_DOWNSTREAM		
	0x16 → SRAM8_ACCESS_CONTESTED		
	0x17 → SRAM8_ACCESS		
	0x18 → SRAM7_STALL_UPSTREAM		
	0x19 → SRAM7_STALL_DOWNSTREAM		
	0x1a → SRAM7_ACCESS_CONTESTED		
	0x1b → SRAM7_ACCESS		
	0x1c → SRAM6_STALL_UPSTREAM		

Bits	Description	Type	Reset
	0x1d → SRAM6_STALL_DOWNSTREAM		
	0x1e → SRAM6_ACCESS_CONTESTED		
	0x1f → SRAM6_ACCESS		
	0x20 → SRAM5_STALL_UPSTREAM		
	0x21 → SRAM5_STALL_DOWNSTREAM		
	0x22 → SRAM5_ACCESS_CONTESTED		
	0x23 → SRAM5_ACCESS		
	0x24 → SRAM4_STALL_UPSTREAM		
	0x25 → SRAM4_STALL_DOWNSTREAM		
	0x26 → SRAM4_ACCESS_CONTESTED		
	0x27 → SRAM4_ACCESS		
	0x28 → SRAM3_STALL_UPSTREAM		
	0x29 → SRAM3_STALL_DOWNSTREAM		
	0x2a → SRAM3_ACCESS_CONTESTED		
	0x2b → SRAM3_ACCESS		
	0x2c → SRAM2_STALL_UPSTREAM		
	0x2d → SRAM2_STALL_DOWNSTREAM		
	0x2e → SRAM2_ACCESS_CONTESTED		
	0x2f → SRAM2_ACCESS		
	0x30 → SRAM1_STALL_UPSTREAM		
	0x31 → SRAM1_STALL_DOWNSTREAM		
	0x32 → SRAM1_ACCESS_CONTESTED		
	0x33 → SRAM1_ACCESS		
	0x34 → SRAM0_STALL_UPSTREAM		
	0x35 → SRAM0_STALL_DOWNSTREAM		
	0x36 → SRAM0_ACCESS_CONTESTED		
	0x37 → SRAM0_ACCESS		
	0x38 → XIP_MAIN1_STALL_UPSTREAM		
	0x39 → XIP_MAIN1_STALL_DOWNSTREAM		
	0x3a → XIP_MAIN1_ACCESS_CONTESTED		
	0x3b → XIP_MAIN1_ACCESS		
	0x3c → XIP_MAIN0_STALL_UPSTREAM		
	0x3d → XIP_MAIN0_STALL_DOWNSTREAM		
	0x3e → XIP_MAIN0_ACCESS_CONTESTED		
	0x3f → XIP_MAIN0_ACCESS		
	0x40 → ROM_STALL_UPSTREAM		

Bits	Description	Type	Reset
	0x41 → ROM_STALL_DOWNSTREAM		
	0x42 → ROM_ACCESS_CONTESTED		
	0x43 → ROM_ACCESS		

BUSCTRL: PERFCTR1 Register

Offset: 0x14

Description

Bus fabric performance counter 1

Table 1326. PERFCTR1 Register

Bits	Description	Type	Reset
31:24	Reserved.	-	-
23:0	Busfabric saturating performance counter 1 Count some event signal from the busfabric arbiters, if PERFCTR_EN is set. Write any value to clear. Select an event to count using PERFSEL1	WC	0x000000

BUSCTRL: PERFSEL1 Register

Offset: 0x18

Description

Bus fabric performance event select for PERFCTR1

Table 1327. PERFSEL1 Register

Bits	Description	Type	Reset
31:7	Reserved.	-	-
6:0	Select an event for PERFCTR1. For each downstream port of the main crossbar, four events are available: ACCESS, an access took place; ACCESS_CONTESTED, an access took place that previously stalled due to contention from other masters; STALL_DOWNSTREAM, count cycles where any master stalled due to a stall on the downstream bus; STALL_UPSTREAM, count cycles where any master stalled for any reason, including contention from other masters.	RW	0x1f
	Enumerated values:		
	0x00 → SIOB_PROC1_STALL_UPSTREAM		
	0x01 → SIOB_PROC1_STALL_DOWNSTREAM		
	0x02 → SIOB_PROC1_ACCESS_CONTESTED		
	0x03 → SIOB_PROC1_ACCESS		
	0x04 → SIOB_PROC0_STALL_UPSTREAM		
	0x05 → SIOB_PROC0_STALL_DOWNSTREAM		
	0x06 → SIOB_PROC0_ACCESS_CONTESTED		
	0x07 → SIOB_PROC0_ACCESS		
	0x08 → APB_STALL_UPSTREAM		
	0x09 → APB_STALL_DOWNSTREAM		
	0x0a → APB_ACCESS_CONTESTED		

Bits	Description	Type	Reset
	0x0b → APB_ACCESS		
	0x0c → FASTPERI_STALL_UPSTREAM		
	0x0d → FASTPERI_STALL_DOWNSTREAM		
	0x0e → FASTPERI_ACCESS_CONTESTED		
	0x0f → FASTPERI_ACCESS		
	0x10 → SRAM9_STALL_UPSTREAM		
	0x11 → SRAM9_STALL_DOWNSTREAM		
	0x12 → SRAM9_ACCESS_CONTESTED		
	0x13 → SRAM9_ACCESS		
	0x14 → SRAM8_STALL_UPSTREAM		
	0x15 → SRAM8_STALL_DOWNSTREAM		
	0x16 → SRAM8_ACCESS_CONTESTED		
	0x17 → SRAM8_ACCESS		
	0x18 → SRAM7_STALL_UPSTREAM		
	0x19 → SRAM7_STALL_DOWNSTREAM		
	0x1a → SRAM7_ACCESS_CONTESTED		
	0x1b → SRAM7_ACCESS		
	0x1c → SRAM6_STALL_UPSTREAM		
	0x1d → SRAM6_STALL_DOWNSTREAM		
	0x1e → SRAM6_ACCESS_CONTESTED		
	0x1f → SRAM6_ACCESS		
	0x20 → SRAM5_STALL_UPSTREAM		
	0x21 → SRAM5_STALL_DOWNSTREAM		
	0x22 → SRAM5_ACCESS_CONTESTED		
	0x23 → SRAM5_ACCESS		
	0x24 → SRAM4_STALL_UPSTREAM		
	0x25 → SRAM4_STALL_DOWNSTREAM		
	0x26 → SRAM4_ACCESS_CONTESTED		
	0x27 → SRAM4_ACCESS		
	0x28 → SRAM3_STALL_UPSTREAM		
	0x29 → SRAM3_STALL_DOWNSTREAM		
	0x2a → SRAM3_ACCESS_CONTESTED		
	0x2b → SRAM3_ACCESS		
	0x2c → SRAM2_STALL_UPSTREAM		
	0x2d → SRAM2_STALL_DOWNSTREAM		
	0x2e → SRAM2_ACCESS_CONTESTED		

Bits	Description	Type	Reset
	0x2f → SRAM2_ACCESS		
	0x30 → SRAM1_STALL_UPSTREAM		
	0x31 → SRAM1_STALL_DOWNSTREAM		
	0x32 → SRAM1_ACCESS_CONTESTED		
	0x33 → SRAM1_ACCESS		
	0x34 → SRAM0_STALL_UPSTREAM		
	0x35 → SRAM0_STALL_DOWNSTREAM		
	0x36 → SRAM0_ACCESS_CONTESTED		
	0x37 → SRAM0_ACCESS		
	0x38 → XIP_MAIN1_STALL_UPSTREAM		
	0x39 → XIP_MAIN1_STALL_DOWNSTREAM		
	0x3a → XIP_MAIN1_ACCESS_CONTESTED		
	0x3b → XIP_MAIN1_ACCESS		
	0x3c → XIP_MAIN0_STALL_UPSTREAM		
	0x3d → XIP_MAIN0_STALL_DOWNSTREAM		
	0x3e → XIP_MAIN0_ACCESS_CONTESTED		
	0x3f → XIP_MAIN0_ACCESS		
	0x40 → ROM_STALL_UPSTREAM		
	0x41 → ROM_STALL_DOWNSTREAM		
	0x42 → ROM_ACCESS_CONTESTED		
	0x43 → ROM_ACCESS		

BUSCTRL: PERFCTR2 Register

Offset: 0x1c

Description

Bus fabric performance counter 2

Table 1328. PERFCTR2 Register

Bits	Description	Type	Reset
31:24	Reserved.	-	-
23:0	Busfabric saturating performance counter 2 Count some event signal from the busfabric arbiters, if PERFCTR_EN is set. Write any value to clear. Select an event to count using PERFSEL2	WC	0x000000

BUSCTRL: PERFSEL2 Register

Offset: 0x20

Description

Bus fabric performance event select for PERFCTR2

Table 1329. PERFSEL2 Register

Bits	Description	Type	Reset
31:7	Reserved.	-	-
6:0	Select an event for PERFCTR2. For each downstream port of the main crossbar, four events are available: ACCESS, an access took place; ACCESS_CONTESTED, an access took place that previously stalled due to contention from other masters; STALL_DOWNSTREAM, count cycles where any master stalled due to a stall on the downstream bus; STALL_UPSTREAM, count cycles where any master stalled for any reason, including contention from other masters.	RW	0x1f
	Enumerated values:		
	0x00 → SIOB_PROC1_STALL_UPSTREAM		
	0x01 → SIOB_PROC1_STALL_DOWNSTREAM		
	0x02 → SIOB_PROC1_ACCESS_CONTESTED		
	0x03 → SIOB_PROC1_ACCESS		
	0x04 → SIOB_PROC0_STALL_UPSTREAM		
	0x05 → SIOB_PROC0_STALL_DOWNSTREAM		
	0x06 → SIOB_PROC0_ACCESS_CONTESTED		
	0x07 → SIOB_PROC0_ACCESS		
	0x08 → APB_STALL_UPSTREAM		
	0x09 → APB_STALL_DOWNSTREAM		
	0x0a → APB_ACCESS_CONTESTED		
	0x0b → APB_ACCESS		
	0x0c → FASTPERI_STALL_UPSTREAM		
	0x0d → FASTPERI_STALL_DOWNSTREAM		
	0x0e → FASTPERI_ACCESS_CONTESTED		
	0x0f → FASTPERI_ACCESS		
	0x10 → SRAM9_STALL_UPSTREAM		
	0x11 → SRAM9_STALL_DOWNSTREAM		
	0x12 → SRAM9_ACCESS_CONTESTED		
	0x13 → SRAM9_ACCESS		
	0x14 → SRAM8_STALL_UPSTREAM		
	0x15 → SRAM8_STALL_DOWNSTREAM		
	0x16 → SRAM8_ACCESS_CONTESTED		
	0x17 → SRAM8_ACCESS		
	0x18 → SRAM7_STALL_UPSTREAM		
	0x19 → SRAM7_STALL_DOWNSTREAM		
	0x1a → SRAM7_ACCESS_CONTESTED		
	0x1b → SRAM7_ACCESS		
	0x1c → SRAM6_STALL_UPSTREAM		

Bits	Description	Type	Reset
	0x1d → SRAM6_STALL_DOWNSTREAM		
	0x1e → SRAM6_ACCESS_CONTESTED		
	0x1f → SRAM6_ACCESS		
	0x20 → SRAM5_STALL_UPSTREAM		
	0x21 → SRAM5_STALL_DOWNSTREAM		
	0x22 → SRAM5_ACCESS_CONTESTED		
	0x23 → SRAM5_ACCESS		
	0x24 → SRAM4_STALL_UPSTREAM		
	0x25 → SRAM4_STALL_DOWNSTREAM		
	0x26 → SRAM4_ACCESS_CONTESTED		
	0x27 → SRAM4_ACCESS		
	0x28 → SRAM3_STALL_UPSTREAM		
	0x29 → SRAM3_STALL_DOWNSTREAM		
	0x2a → SRAM3_ACCESS_CONTESTED		
	0x2b → SRAM3_ACCESS		
	0x2c → SRAM2_STALL_UPSTREAM		
	0x2d → SRAM2_STALL_DOWNSTREAM		
	0x2e → SRAM2_ACCESS_CONTESTED		
	0x2f → SRAM2_ACCESS		
	0x30 → SRAM1_STALL_UPSTREAM		
	0x31 → SRAM1_STALL_DOWNSTREAM		
	0x32 → SRAM1_ACCESS_CONTESTED		
	0x33 → SRAM1_ACCESS		
	0x34 → SRAM0_STALL_UPSTREAM		
	0x35 → SRAM0_STALL_DOWNSTREAM		
	0x36 → SRAM0_ACCESS_CONTESTED		
	0x37 → SRAM0_ACCESS		
	0x38 → XIP_MAIN1_STALL_UPSTREAM		
	0x39 → XIP_MAIN1_STALL_DOWNSTREAM		
	0x3a → XIP_MAIN1_ACCESS_CONTESTED		
	0x3b → XIP_MAIN1_ACCESS		
	0x3c → XIP_MAIN0_STALL_UPSTREAM		
	0x3d → XIP_MAIN0_STALL_DOWNSTREAM		
	0x3e → XIP_MAIN0_ACCESS_CONTESTED		
	0x3f → XIP_MAIN0_ACCESS		
	0x40 → ROM_STALL_UPSTREAM		

Bits	Description	Type	Reset
	0x41 → ROM_STALL_DOWNSTREAM		
	0x42 → ROM_ACCESS_CONTESTED		
	0x43 → ROM_ACCESS		

BUSCTRL: PERFCTR3 Register

Offset: 0x24

Description

Bus fabric performance counter 3

Table 1330. PERFCTR3 Register

Bits	Description	Type	Reset
31:24	Reserved.	-	-
23:0	Busfabric saturating performance counter 3 Count some event signal from the busfabric arbiters, if PERFCTR_EN is set. Write any value to clear. Select an event to count using PERFSEL3	WC	0x000000

BUSCTRL: PERFSEL3 Register

Offset: 0x28

Description

Bus fabric performance event select for PERFCTR3

Table 1331. PERFSEL3 Register

Bits	Description	Type	Reset
31:7	Reserved.	-	-
6:0	Select an event for PERFCTR3. For each downstream port of the main crossbar, four events are available: ACCESS, an access took place; ACCESS_CONTESTED, an access took place that previously stalled due to contention from other masters; STALL_DOWNSTREAM, count cycles where any master stalled due to a stall on the downstream bus; STALL_UPSTREAM, count cycles where any master stalled for any reason, including contention from other masters.	RW	0x1f
	Enumerated values:		
	0x00 → SIOB_PROC1_STALL_UPSTREAM		
	0x01 → SIOB_PROC1_STALL_DOWNSTREAM		
	0x02 → SIOB_PROC1_ACCESS_CONTESTED		
	0x03 → SIOB_PROC1_ACCESS		
	0x04 → SIOB_PROC0_STALL_UPSTREAM		
	0x05 → SIOB_PROC0_STALL_DOWNSTREAM		
	0x06 → SIOB_PROC0_ACCESS_CONTESTED		
	0x07 → SIOB_PROC0_ACCESS		
	0x08 → APB_STALL_UPSTREAM		
	0x09 → APB_STALL_DOWNSTREAM		
	0x0a → APB_ACCESS_CONTESTED		

Bits	Description	Type	Reset
	0x0b → APB_ACCESS		
	0x0c → FASTPERI_STALL_UPSTREAM		
	0x0d → FASTPERI_STALL_DOWNSTREAM		
	0x0e → FASTPERI_ACCESS_CONTESTED		
	0x0f → FASTPERI_ACCESS		
	0x10 → SRAM9_STALL_UPSTREAM		
	0x11 → SRAM9_STALL_DOWNSTREAM		
	0x12 → SRAM9_ACCESS_CONTESTED		
	0x13 → SRAM9_ACCESS		
	0x14 → SRAM8_STALL_UPSTREAM		
	0x15 → SRAM8_STALL_DOWNSTREAM		
	0x16 → SRAM8_ACCESS_CONTESTED		
	0x17 → SRAM8_ACCESS		
	0x18 → SRAM7_STALL_UPSTREAM		
	0x19 → SRAM7_STALL_DOWNSTREAM		
	0x1a → SRAM7_ACCESS_CONTESTED		
	0x1b → SRAM7_ACCESS		
	0x1c → SRAM6_STALL_UPSTREAM		
	0x1d → SRAM6_STALL_DOWNSTREAM		
	0x1e → SRAM6_ACCESS_CONTESTED		
	0x1f → SRAM6_ACCESS		
	0x20 → SRAM5_STALL_UPSTREAM		
	0x21 → SRAM5_STALL_DOWNSTREAM		
	0x22 → SRAM5_ACCESS_CONTESTED		
	0x23 → SRAM5_ACCESS		
	0x24 → SRAM4_STALL_UPSTREAM		
	0x25 → SRAM4_STALL_DOWNSTREAM		
	0x26 → SRAM4_ACCESS_CONTESTED		
	0x27 → SRAM4_ACCESS		
	0x28 → SRAM3_STALL_UPSTREAM		
	0x29 → SRAM3_STALL_DOWNSTREAM		
	0x2a → SRAM3_ACCESS_CONTESTED		
	0x2b → SRAM3_ACCESS		
	0x2c → SRAM2_STALL_UPSTREAM		
	0x2d → SRAM2_STALL_DOWNSTREAM		
	0x2e → SRAM2_ACCESS_CONTESTED		

Bits	Description	Type	Reset
	0x2f → SRAM2_ACCESS		
	0x30 → SRAM1_STALL_UPSTREAM		
	0x31 → SRAM1_STALL_DOWNSTREAM		
	0x32 → SRAM1_ACCESS_CONTESTED		
	0x33 → SRAM1_ACCESS		
	0x34 → SRAM0_STALL_UPSTREAM		
	0x35 → SRAM0_STALL_DOWNSTREAM		
	0x36 → SRAM0_ACCESS_CONTESTED		
	0x37 → SRAM0_ACCESS		
	0x38 → XIP_MAIN1_STALL_UPSTREAM		
	0x39 → XIP_MAIN1_STALL_DOWNSTREAM		
	0x3a → XIP_MAIN1_ACCESS_CONTESTED		
	0x3b → XIP_MAIN1_ACCESS		
	0x3c → XIP_MAIN0_STALL_UPSTREAM		
	0x3d → XIP_MAIN0_STALL_DOWNSTREAM		
	0x3e → XIP_MAIN0_ACCESS_CONTESTED		
	0x3f → XIP_MAIN0_ACCESS		
	0x40 → ROM_STALL_UPSTREAM		
	0x41 → ROM_STALL_DOWNSTREAM		
	0x42 → ROM_ACCESS_CONTESTED		
	0x43 → ROM_ACCESS		